

AVR

Embedded Systems based on AVR Diploma [AVR Diploma]

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passion for technology!



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Memory

Memory used to **store data**.

There are **two** types of memory:

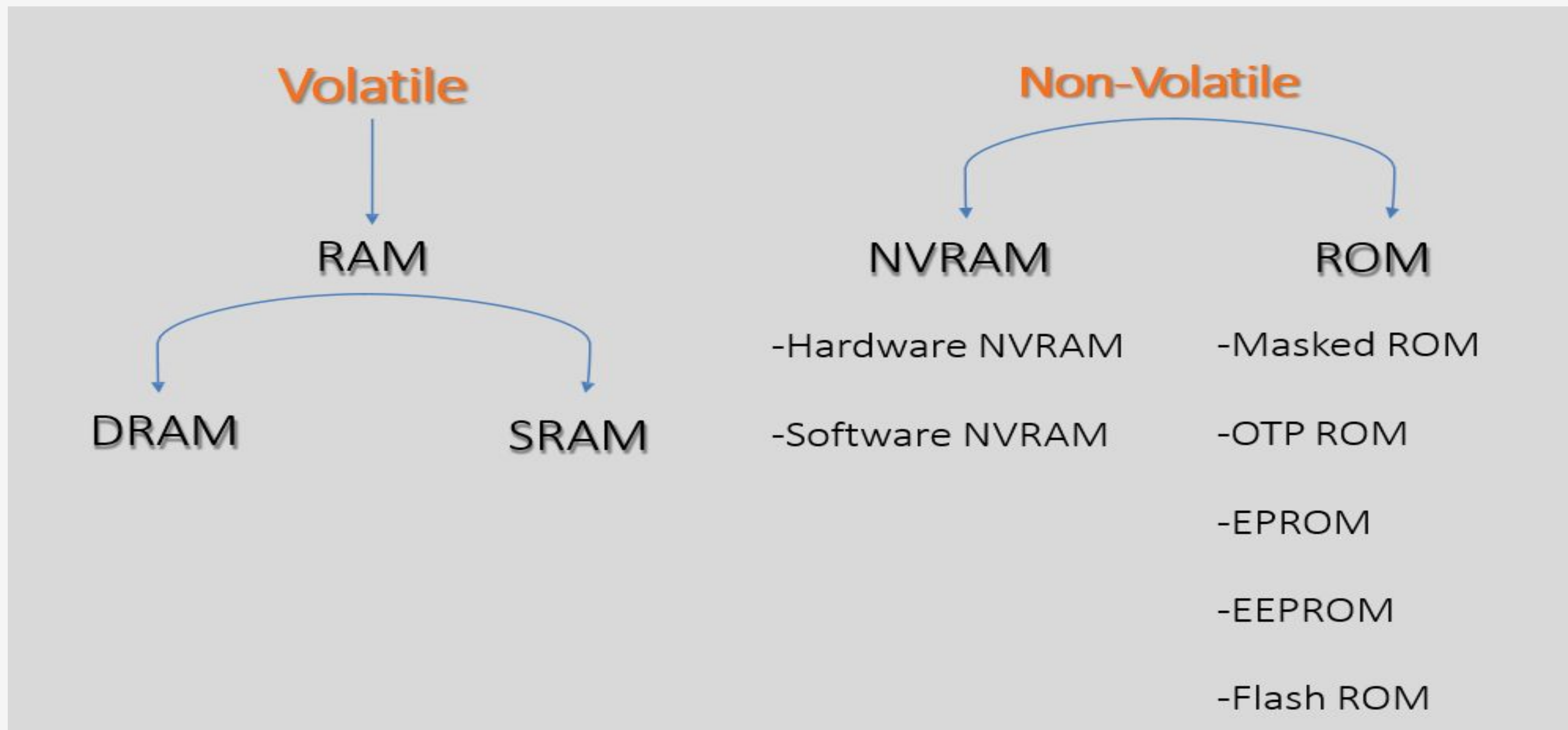
Volatile:

The memory will **lose** its data when the applied voltage **removed** “**power off**”.

Non-Volatile:

The memory will **never lose** its data when the applied voltage **removed** “**power off**”.

Memory Cont.



Volatile Memory

RAM “Random Access Memory”:

It called random access because the time which it takes to reach any address in the memory is constant. (no difference in access time between address 0x0000 and 0xFFFF).

SRAM “Static Random Access Memory”: Based on Transistors.

DRAM “Dynamic Random Access Memory”: Based on Capacitors.

SRAM Vs. DRAM

	SRAM	DRAM
Cost		<i>Lower</i>
Size	<i>Lower</i>	
Performance	<i>Higher</i>	
Power Consumption	<i>Lower</i>	Needs a refresh circuit to recharge capacitors.

So, DRAM is **not applicable** in embedded application because of high **power consumption**.

NON-Volatile Memory

- Masked ROM**: Designed and programmed by manufacturer.
- OTP ROM** “**O**ne **T**ime **P**rogrammable ROM”: Can be programmed only one time.
- EPROM** “**E**rasable **P**rogrammable **R**OM”: Programmed many times and erased by **UV** “**U**ltra **V**iolet”.
- EEPROM** “**E**lectrically **E**rasable **P**rogrammable ROM”: Programmed and erased by electrical signals.
- Flash ROM**

Memory Conclusion

RAM :

for “Run Time” because it is faster than ROM.

ROM :

for “Code” because it doesn’t lose its data after power off.

Note:

Every Embedded system must have **FLASH** and **SRAM** and may have **EEPROM**.

Input output peripherals

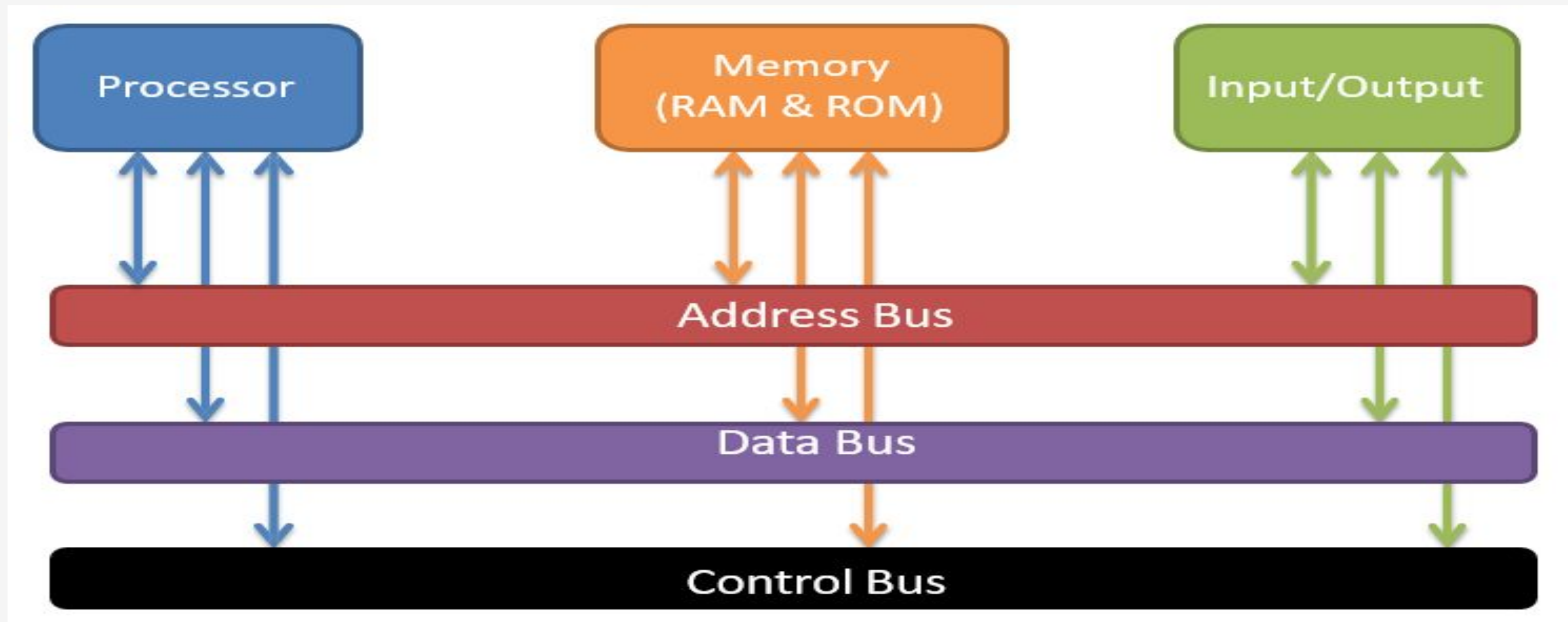
Input and output peripherals as a channel between the user and the processor.

Examples

- 1- Digital Input Output (DIO)
- 2- Analog to Digital Converter(ADC)
- 3-Digital to Analog Converter (DAC)
- 4-Timers and Pulse Width Modulators (PWM)
- 5-Universal Asynchronous Receiver Transmitter (UART)
- 6-Serial Peripheral Interface (SPI)
- 7-Inter Integrated Circuit (I2C)

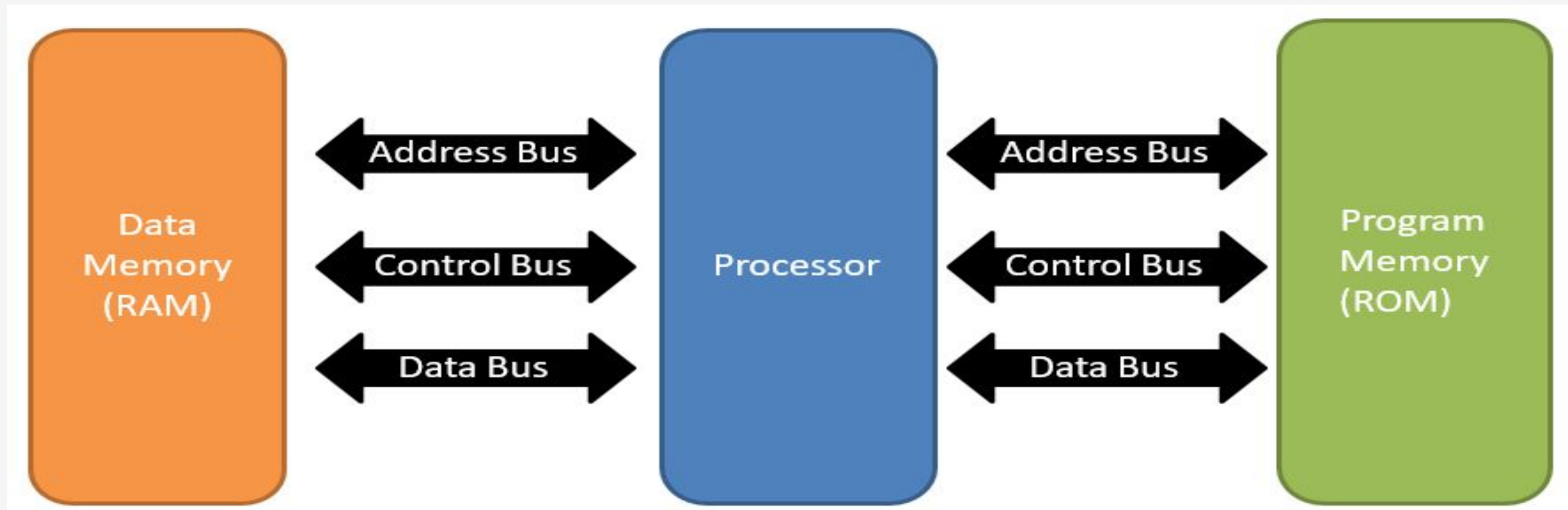
ES Architecture

1- Von Numann Architecture



ES Architecture Cont.

2- Harvard Architecture



Thank You

Do you have any questions?



Contact Us

